MATH317 · NUMERICAL METHODS

تدريب Midterm 471

20/10/2025 · 60 دقيقة · 30 درجة

محارب

01 الأسئلة

حل كل سؤال في المساحة اللي تحته. الحل الكامل في ملف منفصل.

Q1 7 M

Let x = 0.54617 and y = 0.54601. Use four-digit arithmetic to approximate x − y and determine the absolute and relative errors using rounding of numbers.

8 M **Q2**

Given the polynomial $P(x) = 3x^3 - 4x^2 - 12$.

a) Use Horner's algorithm to evaluate $P(-2)$.

b) Using the complete Horner's algorithm, find the Taylor expansion of $P(x)$ about the point $r = 3$.

7 M **Q3**

Determine the single-precision machine representation of the decimal number (-45.21875).

8 M

Q4

Given $f(x) = x^3 - 9x + 1$.

a) Prove that f(x) has exactly one real root α in the interval [2, 4].

b) Find a root of f(x) on [2, 4] by using the bisection method. Terminate the iterations if they exceed nmax = 2 or if the error bound is less than 0.5×10^{-6}.